CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

NIHARIKA PREEJESH (192524506)

STUDENT-TEACHER-SUBJECT CODE DATABASE

Aim:

To implement a Prolog database representing the relationship between students, teachers and subjects.

Objective:

- **To store student information.**
- **To associate teachers with subjects.**
- **To determine which teacher teaches a particular subject.**
- **To determine which subjects are taken by a student.**

Algorithm:

- **Start.**
- **Define student-subject facts.**
- **Define teacher-subject facts.**
- **Create rules to relate students, subjects and teachers.**
- **Query the required relationship.**
- **Display the result.**
- **Stop.**

Flowchart:

```
START
  |
  v
Store Student-Subject Facts
  |
  v
Store Teacher-Subject Facts
  |
  v
Enter Query
  |
  v
Match Facts/Rules
  |
  v
Display Result
  |
  v
STOP
```

Prolog Program:

```
% Student-Teacher-Subject Database

student(rahul, ai).
student(rahul, mathematics).
student(priya, ai).
student(priya, physics).
student(arun, mathematics).
student(arun, physics).

teacher(ramesh, ai).
teacher(suresh, mathematics).
teacher(kavitha, physics).

teaches(Teacher, Student, Subject) :-
    student(Student, Subject),
    teacher(Teacher, Subject).
```

Output:

```
68 ?- teacher_of(Teacher, c_programming).
Teacher = kumar.
```

Result:

Thus, the Prolog program successfully represents and queries student-teacher-subject relationships.